BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Inter, Part I, 2023 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Sum of cube roots of unity is :

(A) 2i
(B) -1
(C) 0
(D) 1

Q2. If nP2 = 30 then n = :

(A) 5
(B) 6
(C) 7
(D) 8

Q3. The modulus of complex number 1 - i sqrt(3) is :

(A) 1 + i sqrt(3)
(B) -1 + i sqrt(3)
(C) 2
(D) 2

Q4. Arithmetic mean between sqrt(2) and 3 sqrt(2) is :

(A) 2 sqrt(2)
(B) sqrt(6)
(C) 3/sqrt(2)
(D) sqrt(2)/2

Q5. If a function f: A -> B is such that Ran f is a subset of B i.e. Ran f != B then f is called:

(A) Into function
(B) Onto function
(C) Injective function
(D) Bijective function

Q6. Partial fractions of (x^2 + 1)/((x + 1)(x - 1)) are of the type:

(A) A/(x+1) + B/(x-1)

(B) 1 - A/(x+1) - B/(x-1)

(C) 1 + A/(x+1) + B/(x-1)

(D) (Ax+B)/(x+1) + C/(x-1)

Q7. Quadratic equation whose roots are 2 and 3:

(A) x^2 - 5x + 6 = 0

(B) x^2 + 5x + 6 = 0

(C) x^2 - 5x - 6 = 0

(D) x^2 + 5x - 6 = 0

Q8. If A is a square matrix of order 3 then |KA| = :

(A) K|A|

(B) K^3|A|

(C) K^2|A|

(D) K^4|A|

Q9. 7th term of the sequence of 2, 6, 11, 17, ... is:

(A) 24

(B) 26

(C) 30

(D) 32

Q10. The trivial solution of homogeneous linear equation is:

(A) (0, 1, 1)

(B) (0, 1, 0)

(C) (1, 0, 0)

(D) (0, 0, 0)

Q11. Domain of the function y = cot x is:

(A) -infinity < x < +infinity

(B) -infinity < x < +infinity, x != (2pi+1)x/2, pi in Z

(C) -1 <= x <= 1

(D) -infinity < x < +infinity, x != n pi, n in Z

Q12. If A and B are overlapping events then P(A union B) = -------- :

(A) P(A) + P(B)

(B) 1 - P(A)

(C) P(A) + P(B) - P(A intersect B)

(D) 1 - P(B)

Q13. The solutions of cosec theta = 2 which lie in [0, 2pi] :

(A) 4pi/3, 5pi/3

(B) 2pi/3, 4pi/3

(C) pi/4, 3pi/4

(D) pi/6, 5pi/6

Q14. cos(pi/2 - beta) = -------- :

(A) -sin beta

(B) sin beta

(C) cos beta

(D) -cos beta

Q15. cos^-1(-x) = :

(A) cos^-1 x

(B) -cos^-1 x

(C) pi - cos^-1 x

(D) 2pi - cos^-1 x

Q16. 2nd term in the expansion of (a/2 + 2/a)^6 is:

(A) a^6/64 (B) 15/4 a^2

(C) -20 (D) 3/8 a^4

Q17. If sin theta = 12/13 and terminal arm is in quad - 1 then cos theta = -------- :

(A) 13/5 (B) -5/13

(C) 5/13 (D) -13/5

Q18. In any triangle with usual notations sin(alpha/2) = :

(A) sqrt((s-a)(s-b)/ab) (B) sqrt((s-b)(s-c)/bc)

(C) sqrt((s-c)(s-a)/ca) (D) sqrt(s(s-c)/ab)

Q19. If n is odd in the expansion of (a + x)^n then number of middle term are:

(A) 2 (B) 3

(C) 4 (D) 1

Q20. In law of cosine if beta = 90 degrees then it reduces to:

(A) b^2 + c^2 = a^2 (B) c^2 + a^2 = b^2

(C) a^2 + b^2 = c^2 (D) c^2 - a^2 = b^2

Essay Type

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Show that z^2 + z-bar^2 is a real number where .....

ii Find the multiplicative inverse of 1 - 2i.

iii Write the descriptive and tubular form of .....

iv Define disjunction.

v If a, b are elements of a group .... Solve ax = b.

vi Find x and y if [[x+3, 1], [-3, 2y-4]] = [[y, 1], [-3, 2x]]

vii Find the cofactors A2 and A22 if A = [[1, -2, 3], [-2, 3, 1], [4, -3, 2]]

viii Without expansion show that det[[2, 3, 1], [1, 1, 0], [2, -5, 5]] = 0

ix Solve the equation 4^(1+x) + 4^(1-x) = 10

x Show that the product of all the three cube roots of unity is unity.

xi If alpha, beta are the roots of ax^2 + bx + c = 0, a != 0, find the value of alpha^2 + beta^2

xii The sum of a positive number and its reciprocal is 26/5. Find the number.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Resolve (7x + 25)/((x + 3)(x + 4)) into partial fractions.

ii If 1/a, 1/b and 1/c are in A.P., show that b = 2ac/(a + c)

iii Sum the series (x - a) + (x + 3a) + ------ to n terms.

iv Find the 5th term of G.P 3, 6, 12, ------

v If 5 is harmonic mean between 2 and b, find b.

vi Find the sum to n terms of the series whose nth term is 3n^2 + n + 1

vii Find the value of n when nP4 : (n-1)P3 = 9:1.

viii How many necklaces can be made from 6 beads of different colours?

ix Find the value of n, when nC10 = (12 x 11)/2

x Verify the statement 1 + 2 + 4 + - - - - + 2^(n-1) = 2^n - 1 for n = 1, 2

xi Calculate by means of binomial theorem (0.97)^3 upto three decimal places.

xii Expand (1 - x)^(1/2) upto three terms.

QQ.4. (Marks 18) Write short answers to any NINE (9) questions:

i Convert 21.256 degrees to the D degrees M'S" form.

ii Verify sin 2theta = 2 sintheta costheta, when theta = 45 degrees

iii Prove the identity costheta + tantheta sintheta = sectheta

iv Prove that sin(180 degrees + alpha) sin(90 degrees + alpha) = -sin alpha cos alpha

v Prove that (cos11 degrees + sin11 degrees)/(cos11 degrees - sin11 degrees) = tan56 degrees

vi Find the value of cos 105 degrees.

vii Find the period of sin(x/5)

viii Find theta, if cos theta = 0.9316

ix Write any two laws of tangents.

x Find the value of R, if a = 13, b = 14, c = 15.

xi Find the value of tan(cos^-1(sqrt(3)/2))

xii Define trigonometric equation. Give one example.

xiii Find the values of theta, satisfying the equation 2sin^2theta - sintheta = 0; theta in [0, 2pi]

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). Prove that det[[b+c, a, a], [b, c+a, b], [c, c, a+b]] = 4abc

QQ.5(b). Solve the equation x^4 - 3x^3 + 4x^2 - 3x + 1 = 0

QQ.6(a). Resolve into partial fractions (5x^2 - 2x + 3)/(x + 2)^3

QQ.6(b). Find the value of n and r when (n-1)C_r : nC_r : (n+1)C_(r+1) = 3:6:11

QQ.7(a). If 1/a, 1/b, 1/c are in G.P., show that the common ratio is +- sqrt(a/c)

QQ.7(b). Show that C(n,1) + C(n,3) + C(n,5) + - - - - - C(n, n-1) = 2^(n-1)

QQ.8(a). Prove that 1/(cosec theta - cot theta) - 1/sin theta = 1/sin theta - 1/(cosec theta + cot theta)

QQ.8(b). Reduce sin^6 theta to an expression involving only function of multiples of theta, raised to first power.

QQ.9(a). Solve the triangle using first law of tangents and then law of sines a = 36.21, b = 42.09, gamma = 40 degrees 29'

QQ.9(b). (Marks 5) Prove that sin^-1(5/13) + sin^-1(7/25) = cos^-1(253/325)

GROUP – II

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

> Note: Inter, Part I, 2023 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Product of all three cube roots of unity is:

(A) i (B) -i

(C) 1 (D) -1

Q2. P(E) represents the probability of an event 'E' and 0 <= P(E) <= 1 for P(E) = 0 the event will be:

(A) Certain
(B) One
(C) Possible
(D) Impossible

Q3. The multiplicative inverse of (1, 0) is:

(A) 0
(B) 1
(C) (1, 0)
(D) (0, 1)

Q4. G.M. between 1 and 16 is:

(A) -5
(B) 4
(C) 6
(D) 8

Q5. Which one of them is unary operation:

(A) Addition
(B) Multiplication
(C) Subtraction
(D) Negation

Q6. Types of rational fractions are:

(A) 1
(B) 2
(C) 3
(D) 4

Q7. Roots of quadratic equation x^2 - 7x + 10 = 0 are:

(A) 2, 5
(B) -2, 5
(C) 2, -5
(D) -2, -5

Q8. If A is a square matrix of order 3 x 3 then |KA| = :

(A) K|A|
(B) K^2|A|
(C) K^3|A|
(D) K^4|A|

Q9. A.M. between x - 3 and x + 5 is:

(A) x + 1
(B) 2x + 1
(C) 2x + 2
(D) 2

Q10. A square matrix A = [a_ij] is called a skew-symmetric if:

(A) A^t = A
(B) A^t = -A
(C) A^t = +- A
(D) A^-1 = A

Q11. The period of tangent function is :

(A) pi/4
(B) pi/2
(C) pi/3
(D) pi

Q12. The probability that an event does not occur, P(complement of E) = :

(A) 1 - P(E)
(B) 1 + P(E)
(C) 2 - P(E)
(D) 2 + P(E)

Q13. Solution of the equation sin x = 1/2 in [0, 2pi] is:

(A) pi/2
(B) pi/6
(C) pi/4
(D) pi/3

Q14. 1 - sin^2 alpha = :

(A) sin 2alpha
(B) sin(alpha/2)
(C) cos 2alpha
(D) cos alpha

Q15. cos^-1(1/2) = :

(A) pi/3
(B) pi/4
(C) pi/6
(D) pi/2

Q16. The total number of terms in the expansion of (a + x)^n is:

(A) n + 2
(B) n + 1
(C) n
(D) n - 1

Q17. 2pi/3 radian in degree is:

(A) 75 degrees
(B) 100 degrees
(C) 110 degrees
(D) 120 degrees

Q18. Delta/s = :

(A) r
(B) r1
(C) r2
(D) r3

Q19. The statement n^2 > n + 3 hold for n = :

(A) 0
(B) 1
(C) 2
(D) 3

Q20. sqrt(s(s-a)(s-b)(s-c)) = :

(A) r
(B) Delta
(C) Delta s
(D) r1

Essay Type

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Show that for all z in C, (z - z-bar)^2 is a real number

ii Simplify (a + bi)^-2

iii Write the power set of {+, -, x, /}

iv Write the converse, inverse of ~p -> q

v Just, convert (A union B)' = A' intersect B' and (A intersect B)' = A' union B' into logical form.

vi If A = [[1, 2], [a, b]] and A^2 = [[0, 0], [0, 0]], find the values of a and b.

vii Solve the equation 2x1 - 2x2 = 5, 5x1 + x2 = 4

viii Define cofactor of an element of matrix

ix Solve the equation x^3 + x^2 + x + 1 = 0

x If alpha, beta are the roots of x^2 - p alpha - beta - c = 0, prove that (1+alpha)(1+beta) = 1 - c

xi Discuss the nature of roots 2x^2 - 5x + 1 = 0

xii Give the statement of factor theorem.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Without finding constants, write (9x - 7)/((x^2 + 1)(x + 3)) into partial fraction form.

ii If a_(n-3) = 2n - 5, find the nth term of A.P.

iii Sum the series 3 + 5 - 7 + 9 + 11 = 13 + 15 + 17 - 19 + ... + 3n terms.

iv If 1/a, 1/b and 1/c are in G.P., then show that common ratio is +- sqrt(a/c)

v If 5 is the H.M. between 2 and b, find the value of b.

vi Write formula for sum(k) and sum(k^2)

vii If nP4 = 11.10.9, then find n.

viii How many signals can be given by 5 flags of different colours using 3 flags at a time?

ix A die is thrown twice. What is the probability that sum of dots shown is either 3 or 11?

x Using binomial theorem, expand (3alpha - x/(3alpha))^4

xi Find middle term in the expansion of (x/2 + 2/x^2)^12

xii Expand (1 - 2x)^(1/3) upto first three terms.

QQ.4. (Marks 18) Write short answers to any NINE (9) questions:

i Define angle in the standard position.

ii If tantheta = -1/3 and the terminal arm of angle is in second quadrant then find sectheta.

iii Find sintheta and costheta for a = 19x/3

iv If alpha, beta, gamma are angles of triangle ABC then prove sin(alpha + beta) = sin gamma

v Without calculator or table, find cos (75 degrees).

vi Prove that tan (45 degrees + A) tan (45 degrees - A) = 1

vii Define period of a trigonometric function.

viii Solve the right triangle ABC in which gamma = 90 degrees, a = 3.28, b = 5.74

ix By using the law of cosine, write the formula of cos alpha and cos beta

x Solve the triangle ABC if beta = 60 degrees, gamma = 15 degrees and b = sqrt(6)

xi Define the principal sin function.

xii Solve the equation sin x = 1/2

xiii Solve the equation sin x + cos x = 0 and find its general solution set.

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 5) If A = [[1, 1+i], [1, -i]] show that A + (A-bar)^t is skew Hermitian.

QQ.5(b). (Marks 5) When x^4 + 2x^3 + kx^2 + 3 is divided by x - 2 and remainder is 1, find the value of k.

QQ.6(a). (Marks 5) Resolve into partial fraction 1/((x - 1)^2 (x + 1))

QQ.6(b). (Marks 5) Prove that nC_r + nC_(r-1) = (n+1)C_r

QQ.7(a). (Marks 5) Find 'n' so that (a^(n+1) + b^(n+1))/(a^n + b^n) may be H.M between a and b.

QQ.7(b). (Marks 5) Find (2n + 1)th term from the end in expansion of (x - 1/(2x))^(2n)

QQ.8(a). (Marks 5) If tan theta = 1/sqrt(3) and the terminal arm of the angle is not in the III quad, find the value of (cosec^2 theta - sec^2 theta)/(cosec^2 theta + sec^2 theta)

QQ.8(b). (Marks 5) Prove that cos 20 degrees cos 40 degrees cos 60 degrees cos 80 degrees = 1/16

QQ.9(a). (Marks 5) Solve the triangle ABC if a = 7, b = 3, gamma = 38 degrees 13'

QQ.9(b). (Marks 5) Prove that tan^-1(3/4) + tan^-1(1/5) - tan^-1(8/19) = x/4